

Mahmoud Alyosify

AI & Machine Learning Engineer | LLM Systems, Applied ML & Model Deployment | MSc Artificial Intelligence, Queen's University
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🌐 [linkedin.com/in/mahmoudalyosify](https://www.linkedin.com/in/mahmoudalyosify) | 🐙 github.com/MahmoudAlyosify | 🌐 Portfolio | 🏠 Hugging Face | 🏆 Kaggle

PROFESSIONAL SUMMARY

AI & Machine Learning Engineer who builds intelligent systems end to end — from research question to a deployed, measured model. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**, with a master's research project on **training-free, decoding-time efficiency for large language models**. Core strengths in **LLM fine-tuning and inference optimisation**, **end-to-end ML pipelines** and **cloud deployment on AWS**, backed by rigorous, reproducible evaluation. Prior engineering experience on national-scale toll infrastructure, plus five years teaching AI and computer science to **10,000+ learners (700K+ views)**. Open to AI/ML engineering roles internationally.

EDUCATION

- MSc in Artificial Intelligence** Sep 2025 – Nov 2026
Queen's University, Kingston, Ontario, Canada Cum GPA: 4.30 / 4.30
- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
 - **Master's Research Project (CISC 898):** *Verbosity-Aware Decoding for Token-Efficient Language Model Inference* — Supervisor: Dr. Rahatara Ferdousi.
 - Coursework: **Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines.**
- B.Sc. Computer & Information Science (Bioinformatics)** Sep 2019 – Jul 2023
Assiut University, Egypt GPA: 3.53 / 4.00
- Graduation Project: **Vitalism Solution** — contactless vital-sign estimation from video via rPPG. **Grade A+; Semi-Finalist, Microsoft Imagine Cup 2023.**

TECHNICAL SKILLS

AI & LLM Systems	Large Language Models, Fine-Tuning (LoRA, QLoRA, PEFT), RAG, Transformers, Hugging Face, LangChain, Prompt Engineering, Multi-Agent Systems, Agentic Workflows
ML & Deep Learning	PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CNNs, Computer Vision, NLP, Self-Supervised & Contrastive Learning, Reinforcement Learning (PPO, DQN), Optuna, SHAP
Data & MLOps	Pandas, NumPy, PySpark, ETL Pipelines, Feature Engineering, AWS, Docker, CI/CD, ONNX, FastAPI, Streamlit, FAISS, ChromaDB, MySQL, MongoDB, Git/GitHub, Linux
Evaluation	Benchmarking (MT-Bench, XSum, SQuAD-v2), BERTScore, ROUGE, LLM-as-Judge, Statistical Testing, Ablation Studies, Reproducible Experiment Design
Programming	Python, SQL, C++, C, C#, Go, JavaScript, MATLAB, Bash

PROFESSIONAL EXPERIENCE

- Machine Learning Instructor** Jul 2025 – Sep 2025
National Telecommunication Institute (NTI) Cairo, Egypt
- Delivered a **90-hour** hands-on ML program for engineering and CS students covering supervised learning, PCA and neural networks, with practical lab sessions throughout.
- AI in Cybersecurity Intern** Jun 2025 – Jul 2025
e& Egypt — via NTI (On-the-Job Training) Cairo, Egypt
- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.
- Systems Engineer (Military Service)** Feb 2024 – Mar 2025
National Company for Roads Building & Development Cairo, Egypt
- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**.
 - Co-developed an internal data-management and reporting desktop application in **C#, .NET, Entity Framework, SQL Server** and **Crystal Reports**; supported and troubleshooted legacy applications to minimise operational downtime.
- Online Instructor — AI, Algorithms & Data Science** 2020 – Present
Udemy & YouTube Remote
- Built and taught Arabic-language technical curricula (discrete mathematics, algorithms, probability & statistics, graph data mining) reaching **10,000+ learners** and **700K+ YouTube views**.

SELECTED PROJECTS

- Verbosity-Aware Decoding for Token-Efficient Language Model Inference** 2026 – Present
MSc Research Project (CISC 898), Queen's University · PyTorch · Hugging Face · AWS GPU
- Designed a **training-free LogitsProcessor** that cuts generated tokens at **decoding time** using stop-token, n-gram-repetition and

- semantic-redundancy signals under an **entropy-driven gate** — no retraining, no architecture change, no KV-cache overhead.
- Validated across **three open-weight model families** (Mistral-7B, Llama-3.1-8B, Qwen2.5-3B) on **MT-Bench**, **XSum** and **SQuAD-v2**, measuring token reduction, latency (TTFT / inter-token) and **GPU energy** while proving semantic preservation through paired non-inferiority testing.

HORUS Sentinel — Autonomous Multi-Agent OSINT & Threat-Intelligence Platform
Python · Multi-Agent Systems · Fine-Tuned LLM · Knowledge Graphs · Self-Hosted Inference

2026

- Architected an autonomous intelligence analyst: a **swarm of specialised agents** continuously collects open-source intelligence and correlates findings into a **living knowledge graph**, over which a self-hosted fine-tuned LLM produces **prioritised, evidence-backed reports**.

SimCLR-Vision-SSL — Self-Supervised Contrastive Representation Learning
PyTorch · SimCLR · SupCon · ONNX · FAISS

2026

- Built a self-supervised vision pipeline with a **42-experiment augmentation sweep**, a semi-supervised **SupCon** extension and deployment-ready **ONNX/FAISS** retrieval, reaching **84.30% top-1 accuracy**.

RL-Cloud-Autoscaler — Cloud Resource Provisioning via Deep Reinforcement Learning
Python · PPO · DQN · Gymnasium · Cost Optimisation

2026

- Built a **custom Gymnasium environment** and trained **PPO** and **DQN** agents to optimise the **cost-latency trade-off** under fluctuating demand, benchmarked against static and threshold-based autoscaling baselines.

Additional Projects

- Horus-OSINT** — fine-tuned **Meta-Llama-3-8B** on **159,826** GTD/GDELT records and deployed it on **AWS** as an interactive threat-intelligence assistant.
- Automated Multimodal Agent** — agentic workflow converting PDFs into narrated PowerPoint decks end to end.
- SCRAP** — supervised ML pipeline predicting satellite collision risk **48+ hours** before closest approach.
- AudioShield** — CNN-based deepfake audio detection served through a Streamlit application.
- Also:** Human vs. Agentic Pull Requests (GitHub mining) · Vitalism Solution (rPPG) · Diabetes Onset Prediction · ATM Transaction Analysis · E-Commerce Web App · Mini-FAT File System.

CERTIFICATIONS

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| AWS Academy — ML Foundations, ML for NLP, Data Engineering | Machine Learning Specialization — DeepLearning.AI |
| HCIP – AI (Huawei) | Microsoft Certified: Data Analyst Associate |
| AI in Cybersecurity — NTI (420 hrs) | ML Internship — ITIDA / Egypt Makes Electronics (200 hrs) |
| Generative AI & Prompt Engineering — IBM / Coursera | Full Stack Diploma — .NET & Angular (Route Academy) |

ACHIEVEMENTS & AWARDS

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| 1st Place — Smart Cities Hackathon 2022 | Semi-Finalist — Microsoft Imagine Cup 2023 |
| Finalist — ISEIC 2023 | Ideal Student Award — Assiut University, 2022–2023 |

LEADERSHIP & LANGUAGES

- Secretary**, Higher Scientific & Technological Committee, Student Union, Assiut University · **Ambassador**, Information Technology Institute (ITI), Assiut · **Volunteer Team Leader**, USAID University Center for Career Development.
 - Languages:** Arabic (Native) · English (Professional Working Proficiency) · French (Basic).
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